

Fine-tuning LLaMA 3.2-1B for Buyer Intent Classification

Group Name: 1(Group 18)

Team Members and Contributions

Victoria CHENG

- Implemented zero-shot and few-shot experiments
- Set up the model access and data loading

Rui TAO

- Designed and implemented LoRA fine-tuning experiments
 - Created prompt templates for different evaluation approaches
 - Optimized training parameters and evaluation strategies

Model Performance Comparison

- LLaMA 3.2-1B showed mixed zero-shot performance (Macro F1=0.5509) but after fine-tuning improved dramatically (Macro F1=0.8689).
- Previously challenging categories like "Offensive Intent" improved from F1=0.0548 to F1=0.8430 after fine-tuning.

LoRA Configuration & Hyperparameters

- Higher rank LoRA configuration ($r=16$, $\alpha=32$) targeting attention layers significantly outperformed lower rank settings (Macro F1 of 0.8826 vs 0.2419).
- Targeting attention mechanism layers (q_proj , k_proj , v_proj , o_proj) provided better capacity for learning intent classification.

Hyperparameter Tuning & Training Strategy

- Batch size 4 with gradient accumulation steps of 8 and moderate learning rate ($3e-5$) provided optimal training stability.
- Three epochs were sufficient to avoid overfitting, with evaluation every 200 steps.

- The direct prompt template consistently outperformed step-by-step reasoning across all configurations.

Future Improvements & Lessons Learned

- Expanding the training dataset with more diverse examples and implementing specialized loss functions for class imbalance could further improve performance.
- LoRA fine-tuning achieved excellent results while training only 0.5% of parameters, making efficient adaptation of LLMs practical even with limited computational resources.